Unifying Theory of Dynamic Aether Mechanics: Derived Values Reference Catalog

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Abstract

This reference catalogs roughly seventy-five principal quantitative derivations of the Dynamic Aether Mechanics framework, organized by the type of physical quantity each derivation yields. Each entry lists the result, a brief description, a high-level sketch of how the value is derived, and the originating paper and section. It is a navigation aid for the multi-paper framework, not a self-contained treatment; complete reasoning lives in the companion papers.

1. Reading the Table

Each category presents a small set of *headline* derivations, marked with a bold leading phrase in the description column, followed by their *sub-derivations* marked with the glyph ▹. Headline rows correspond to keystone quantities the framework predicts—force constants, masses, wave speeds, characteristic radii, and so on—while sub-rows record intermediate expressions, observational matches, or related forms that support the headline. The Source column uses the shorthand "P$\langle n \rangle$ §$\langle$sec$\rangle$": for example, *P3 §2.1* refers to Paper 3, Section 2.1. Sidecar papers retain their filename markers: *P6a*, *P6b*, *P7a*. Where a result appears in more than one paper, sources are separated by a semicolon.

The four fundamental Aether parameters that recur throughout are: the bulk modulus K_A, the rest density ρ_A, the atomic mass m_A, and the binding-cycle asymmetry ε. The wave speed $c = \sqrt{K_A/\rho_A}$ and healing length $\xi = \hbar/(m_A c\sqrt{2})$ are derived from this set.

2. Force Constants & Couplings

These rows establish that gravity, electromagnetism, the strong coupling, and the fine-structure constant all reduce to combinations of K_A, ρ_A, and the binding-cycle asymmetry ε.

Derived Math	Description	Derivation	Source
$G \;\propto\; \varepsilon\,\rho_A\,\psi^2$	**Gravitational constant** as the binding-cycle asymmetry ε acting on the local condensate amplitude ψ.	Average the net inward force over one bind/unbind cycle of a transient virtual pair; only the fraction ε that fails to recycle remains.	*P3 §1.3; P1 §2*
▹ $\alpha_G \;=\; G\,m_e^2/(\hbar c) \sim 10^{-42}$	Dimensionless gravitational coupling; the 10^{-42} hierarchy is the binding-cycle asymmetry ε itself.	Form the dimensionless coupling by combining G with m_e, $\hbar$, c; the magnitude collapses to ε.	*P1 §2; P3 §1.3*
$k \;=\; \dfrac{\pi\,K_A\,\hbar^4}{16\,m_e^2\,m_A^2\,c^4\,e^2}$	**Coulomb constant** from the electron-ring axial oscillation acting as a phase-locked monopole radiator.	Apply Bjerknes' secondary-force formula to two oscillating monopoles, substitute the ring's piston area and frequency, then close the amplitude at $A = \xi$.	*P7a §11.4*
▹ $K_C \equiv \rho_A/m_A^2 \approx 7.085 \times 10^{65}\ \mathrm{kg^{-1}\,m^{-3}}$	Single combined-parameter constraint k must satisfy; tightens the allowed (ρ_A, m_A) zone.	Invert the Coulomb-constant formula using CODATA k to extract the combination ρ_A/m_A^2.	*P7a §11.7*
▹ $k_{\text{pred}} \approx 8.6 \times 10^9\ \mathrm{N\,m^2/C^2}$	Direct numerical evaluation against measured 8.99×10^9; agreement within 4% with no free parameters.	Plug the parameter midpoint $\rho_A = 10^{12}$ kg/m^3, $m_A = 680$ MeV/c^2 into the Coulomb formula and evaluate.	*P7a §11.5*
$\mu_0 \;=\; \dfrac{4\pi\,k}{c^2}$	**Vacuum permeability** as the flow-coupling counterpart of the static electric coupling, fixed by propagation speed.	Require the EM wave speed to equal c; this fixes μ_0 in terms of k.	*P4 §3.3*

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Derived Math	Description	Derivation	Source
$\alpha = \dfrac{e^2}{4\pi\varepsilon_0 \hbar c}$	**Fine-structure constant** as the strength of the electron's coherent axial wave field at the quantum scale.	Combine the derived k with $\hbar$ and c to form the dimensionless coupling that controls atomic spectra.	*P1 §2; P4 §3.3*
$\sigma \approx 1\ \text{GeV/fm} = 1.6 \times 10^5\ \text{J/m}$	**QCD string tension** as linear energy per unit length of a compressed-Aether bridge between quarks.	Integrate the bridge elastic-energy density over a cross-section of width ξ, using the logarithmic equation of state at compression $X \sim 10^7$.	*P5 §1.1*
▹ $f = 1 - 1/n^2$	Fresnel drag coefficient for light in a moving refractive medium.	Treat the excess density $(n^2 - 1)\rho_A$ as advected with the bulk while the ambient ρ_A stays still; superpose the two wave speeds.	*P2 §2.5*

3. Wave Speeds & Velocities

The Aether's bulk modulus and density set a single propagation speed for all fundamental waves; gravitational inflow and bridge-internal modes derive from this same medium.

Derived Math	Description	Derivation	Source
$c = \sqrt{K_A/\rho_A}$	**Speed of light** as the longitudinal wave speed in the Aether medium.	Solve the wave equation for a linear elastic continuum with stiffness K_A and density ρ_A.	*P1 §2; P2 §1*

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Derived Math	Description	Derivation	Source
$v_{\text{in}}(r) = \sqrt{2GM/r}$	**Gravitational inflow velocity** of Aether into a mass; equals the escape velocity by energy conservation.	Set the kinetic energy of a free-falling Aether parcel equal to its gravitational potential energy at infinity.	*P2 §1.3; P3 §1.1*
$c_1^2 = c^2 + \dfrac{T\,s^2\,\rho_s}{c_v\,\rho_n}$	**First-sound speed** in the two-fluid Aether; the temperature-dependent correction is the dark-matter contribution to orbital velocities.	Compute the adiabatic sound speed in the Landau two-fluid model, retaining the thermal-coupling term between superfluid and normal components.	*P6 §1.3*
$\triangleright\, c_{\text{phonon}} = c/\sqrt{X} \approx 10^{-4}\,c \approx 3 \times 10^4$ m/s	Longitudinal sound speed inside a compressed-Aether bridge; K_A is density-independent so $c \propto \rho^{-1/2}$.	Apply $\sqrt{K/\rho}$ inside the bridge: K stays at K_A but ρ increases by factor X, so the speed drops by $\sqrt{X}$.	*P5 §5.3*
$\triangleright\, c_{\text{spin}} = c$	Transverse spin-wave speed inside the bridge; density-independent because spin stiffness $\propto \rho$ cancels.	For transverse spin waves the stiffness scales with density; the ratio K/ρ stays at c^2.	*P5 §5.3*
$\triangleright\, v(\xi) = c/\sqrt{2} \approx 0.707\,c$	Azimuthal flow speed at the healing-length boundary of an electron vortex ring; exact and m_A-independent.	Evaluate the irrotational vortex velocity $\kappa_e/(2\pi r)$ at $r = \xi = \hbar/(m_A c\sqrt{2})$; the m_A's cancel.	*P7a §3.1*
$\triangleright\, v_{\text{perm}} \sim c/\sqrt{\pi} \approx 0.56\,c$	Dynamic-Casimir permanence-threshold velocity: above this, vortex material remains permanently bound.	Analogize to Schwinger pair production: the boundary flow speed at which the DCE binding becomes permanent rather than transient.	*P7a §6.3*

4. Lengths, Radii & Wavelengths

Characteristic length scales emerge from balances between inflow, healing, and quantization.

Derived Math	Description	Derivation	Source
$r_s = \dfrac{2GM}{c^2}$	**Schwarzschild radius** as the sonic horizon at which inflow speed equals the wave speed.	Set $v_{\text{in}}(r) = \sqrt{2GM/r}$ equal to c and solve for r.	*P2 §1.4; P3 §7*
$\xi = \dfrac{\hbar}{m_A\, c\sqrt{2}}$	**Aether healing length** as the intrinsic thickness of the core transition zone where density and phase rejoin ambient.	Balance the condensate's kinetic gradient energy against its interaction energy at the boundary of a vortex core.	*P7a §11.4*
$r_{\text{loop}} = \lambda\!\!\!^{-}_C = \hbar/(m_e c)$	**Electron-ring major radius** fixed to the reduced Compton wavelength by self-consistency of the Donnelly mass equation.	Iterate the Donnelly mass equation for self-consistency: the loop radius converges to the reduced Compton wavelength.	*P7a §5.1*
$\lambda_{\text{weak}} \sim \hbar/(m_W c) \approx 10^{-18}$ m	**Weak interaction range** as the inverse W-boson mass: the propagation distance of a virtual gapped excitation.	Apply the uncertainty principle to a virtual W boson of mass m_W; the propagation distance is $\hbar/(m_W c)$.	*P5 §1.9*
▹ $r_{\text{break}} \approx 1.2$ fm	String-breaking distance: quark separation where stored bridge energy reaches the pair-creation threshold.	Set bridge energy $\sigma\, r$ equal to the rest energy of a quark-antiquark pair; solve for r.	*P5 §1.2*
▹ $d_{\text{bridge}} \approx 0.17$ fm	Effective bridge core diameter from thermal disruption at $T_{\text{deconf}} \approx 170$ MeV.	Equate the thermal energy at deconfinement to the bridge energy spanning one diameter: $k_B T_{\text{deconf}} = \sigma\, d_{\text{bridge}}$.	*P5 §1.7*

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Derived Math	Description	Derivation	Source
▹ $r_{\mathrm{eq}} = \kappa_e/(2\pi\, v_{\mathrm{perm}})$	Equilibrium core radius: azimuthal flow at the boundary equals the permanence threshold.	Solve $v(r) = \kappa_e/(2\pi r) = v_{\mathrm{perm}}$: the radius at which boundary flow first reaches the threshold.	*P7a §3.3*
▹ $\lambda\!\!\!^{-}_C = \hbar/(m_e c)$	Reduced Compton wavelength; emergent length scale at which axial oscillation, ring radius, and zitterbewegung all coincide.	Apply the uncertainty principle to a particle of mass m_e at the relativistic momentum scale $m_e c$.	*P7 §1; P7a §5.1*

5. Masses

Particle masses emerge as kinetic energies of vortex flow, bridge tension integrals, or condensate gap excitations.

Derived Math	Description	Derivation	Source
$m_e\, c^2 \approx E_{\mathrm{KE}}^{\mathrm{Donnelly}}\,(1 - 0.017 + f_{\mathrm{core}})$	**Electron rest mass** as Donnelly vortex KE with Bernoulli density-deficit and core-energy corrections; converges on 0.511 MeV.	Start with the Donnelly vortex KE, subtract 1.7% for the Bernoulli density deficit, then add the f_{core} contribution of bound ring material.	*P7a §4.3; P7a §5.2*
$m_p\, c^2 \approx 3\sigma L_{\mathrm{arm}} + m_q c^2 + E_J \approx 938$ MeV	**Proton rest mass** as the integrated tension of three bridge arms (~0.3 fm each) plus quark masses and Y-junction binding.	Sum three bridge-arm energies $\sigma\, L_{\mathrm{arm}}$, three current-quark masses, and the Y-junction binding contribution.	*P5 §1.5*

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Derived Math	Description	Derivation	Source
$m_W \approx 80.4$ GeV, $m_Z \approx 91.2$ GeV	**Weak boson masses** as gapped excitations of the spinor condensate at the topology-changing energy scale.	Identify the W and Z as the energy cost of a vortex-topology transition in the spinor condensate, set by condensate stiffness.	*P5 §1.9*
$m_A \in [628, 689]\ \mathrm{MeV}/c^2$	**Aether atomic mass** narrowed from the joint (ρ_A, m_A) posterior by both the electron-mass and Coulomb constraints.	Intersect the electron-mass and Coulomb-constant constraint curves in the (ρ_A, m_A) plane; the overlap is the allowed range.	*P7a §11.7*
▹ $m_{\mathrm{ring}}c^2 = f_{\mathrm{core}}\, m_e c^2$, $f_{\mathrm{core}} \approx 5$–$14\%$	Bound-core ring material mass: small fraction of electron rest energy carried by DCE-trapped condensate.	Subtract the Bernoulli-corrected Donnelly KE from $m_e c^2$; the remainder is the ring-material contribution.	*P7a §5.2*
▹ $m_q^* \sim \sqrt{\sigma\, \hbar c/\pi} \approx 250$ MeV	Chiral/potential crossover quark mass: Schwinger exponent ~ 1 separates pion-cloud from heavy-quark physics.	Set the Schwinger pair-creation exponent $\pi m_q^2/(\sigma\, \hbar c)$ equal to unity and solve for m_q.	*P5 §1.8*
▹ $m_e^2 = \dfrac{\pi^2 \rho_A\, \hbar^3}{2\, m_A^2\, c^3}\left[\ln\left(\dfrac{8\sqrt{2}\, m_A}{m_e}\right) - \frac{1}{2}\right]$	Self-consistent transcendental equation for m_e after $r_{\mathrm{loop}} = \lambda\!\!\!^{-}_C$ substitution.	Substitute $r_{\mathrm{loop}} = \hbar/(m_e c)$ into the Donnelly formula; the result is a transcendental equation that closes on m_e.	*P7a §5.1*
▹ $\delta E_{\mathrm{KE}}/E_{\mathrm{KE}} \approx -1/[8\ln(r_{\mathrm{loop}}/\xi)] \approx -1.7\%$	Bernoulli density-deficit correction to the bare Donnelly kinetic energy.	Integrate the Bernoulli density deficit weighted by kinetic-energy density over the vortex volume.	*P7a §4.2*

6. Energies

Rest energy, kinetic energy, thermal radiation, and quantum corrections to the bridge potential all derive from the same elastic-compression bookkeeping.

Derived Math	Description	Derivation	Source
$E = m\,c^2$	**Mass–energy equivalence** as the total compression-wave energy released when bound matter dissolves into the Aether.	Compute the elastic energy stored when a mass dissolves into a compression wave; equipartition sums kinetic and elastic parts to mc^2.	*P2 §1.5*
▹ $E_k = E_p = \frac{1}{2}mc^2$	Equipartition of kinetic and elastic energy in a free Aether compression wave.	For a progressive wave in a linear elastic medium, kinetic and elastic energy densities are equal; each carries half of mc^2.	*P2 §1.5*
$\mathrm{KE}_{\mathrm{rel}} = (\gamma - 1)\,mc^2$	**Relativistic kinetic energy** from the work integral $\int v\,dp$ with $p = \gamma mv$; recovers $\frac{1}{2}mv^2$ at low v.	Integrate $\int v\,dp$ along an acceleration path using $p = \gamma mv$, then subtract the rest energy mc^2.	*P2 §1.6*
$T_H = \dfrac{\hbar\,c^3}{8\pi\,G\,M\,k_B}$	**Hawking temperature** from the inflow-velocity gradient at the sonic horizon.	Apply the surface-gravity formula $T = \hbar\,\kappa/(2\pi k_B c)$ with $\kappa = \lvert dv_{\mathrm{in}}/dr\rvert_{r_s} = c^3/(4GM)$.	*P6 §2.1*
▹ $\Delta V(L) = -\dfrac{\pi\,\hbar\,c}{12\,L}$	Lüscher quantum correction to the quark–antiquark linear potential from transverse zero-point modes.	Sum the Bogoliubov zero-point energy of the two transverse-polarization modes of a bridge of length L.	*P5 §1.8*

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Derived Math	Description	Derivation	Source
▹ $\varepsilon_{\text{compress}} = K_A\,[\,X - \ln X - 1\,] \approx K_A X$ for $X \gg 1$	Compression energy density of an Aether bridge at density-enhancement factor X.	Integrate $P\,dV$ along the logarithmic equation of state $P = K_A \ln(\rho/\rho_0)$ from ρ_0 up to $X\rho_0$.	*P5 §5.3*
▹ $\dot{\varepsilon} = 9\,\zeta_n\,H^2$	Bulk-viscous dissipation rate: energy injected into the dark-energy sector by Hubble expansion.	Apply the bulk-viscosity dissipation formula to a pure-dilation flow with trace velocity gradient $3H$, giving $(3H)^2\zeta_n$.	*P6 §1.2; P6a §3.1*

7. Time & Lorentz Factors

The relativistic dilations all arise from the finite propagation speed of binding waves in the Aether.

Derived Math	Description	Derivation	Source
$\gamma = 1/\sqrt{1 - v^2/c^2}$	**Lorentz factor** from the Pythagorean path-length difference between helical and straight wave paths through the medium.	Compute the ratio of helical to straight wave-path lengths for a system moving at v; the result is γ.	*P2 §1.2*
▹ $L_{\text{moving}} = L_{\text{rest}}/\gamma$	Physical length contraction along the direction of motion: retarded-field equilibrium contracts by $1/\gamma$.	Solve for the equilibrium length of a bound system whose binding forces propagate at c while the system moves through the medium.	*P2 §1.1*
$\dfrac{\Delta t}{t} = \dfrac{v_{\text{in}}^2}{2c^2} = \dfrac{GM}{r\,c^2}$	**Gravitational time dilation** as the velocity time dilation of a static observer in flowing Aether.	Apply the velocity time-dilation $v^2/(2c^2)$ to a static observer with Aether flowing past at $v_{\text{in}} = \sqrt{2GM/r}$.	*P2 §1.3; P3 §1.1*

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Derived Math	Description	Derivation	Source
▹ $\delta\rho/\rho_A \approx v^2/(2c^2)$	Velocity-induced density perturbation of an object moving through the Aether; cause of length contraction in the dynamic frame.	Apply Bernoulli's equation to an object moving at v through the Aether; the kinetic energy density appears as excess pressure / compression.	*P2 §1.1*
▹ $t_{\parallel}/t_{\perp} = \gamma$	Round-trip signal-time asymmetry between parallel and perpendicular paths; cancelled by length contraction in Michelson–Morley.	Compute round-trip times: Doppler-asymmetric upstream/downstream for parallel paths; Pythagorean for perpendicular paths.	*P2 §1.1*

8. Field Falloff & Force Laws

Inverse-power laws emerge from spherical-wave spreading; relative velocities generate magnetic and gravitomagnetic corrections governed universally by v^2/c^2.

Derived Math	Description	Derivation	Source
$F_C = \dfrac{k\, q_1\, q_2}{r^2}$	**Coulomb's law** as the secondary Bjerknes force between coherent monopole axial-oscillation radiators.	Apply Bjerknes' secondary-force formula to two coherent acoustic monopoles; wave intensity falls as $1/r^2$.	*P4 §2.5*
$F_G = \dfrac{G\, m_1\, m_2}{r^2}$	**Newtonian gravity** from the dissipating residual binding-pressure wave.	The residual binding-pressure pulse from one mass spreads as $1/r^2$ and acts on the bound matter of the second mass.	*P3 §1.3*

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Derived Math	**Description**	**Derivation**	**Source**
$F_{\text{mag}}/F_{\text{elec}} = v_1 v_2/c^2$	**Magnetic force ratio** between moving charges from two first-order Doppler factors on coherent wavefronts.	Apply one first-order Doppler correction per moving charge to the coherent wavefronts; the product is $v_1 v_2/c^2$.	*P4 §1.4*
$\theta_{\text{lens}} = \dfrac{4\,GM}{r_0\,c^2}$	**Gravitational lensing angle** decomposed into equal advection and time-delay halves.	Integrate the photon trajectory through the inflowing Aether; sum the transverse-advection bend and the time-delay asymmetric bend.	*P3 §2.1*
$\triangleright\, B \propto 1/r^3$	Magnetic dipole falloff from partial cancellation of opposing poles in the flow-angle field.	Subtract the contributions of two opposing poles separated by d; the leading $1/r^2$ terms cancel, leaving $1/r^3$.	*P4 §1.3*
$\triangleright\, \nabla \cdot \mathbf{B} \equiv 0$	Vanishing divergence: magnetic monopoles forbidden by the vortex topology of the flow-angle field.	$\mathbf{B}$ is the curl of a flow field; the divergence of any curl vanishes by vector-calculus identity.	*P4 §1.2*
$\triangleright\, \theta_{\text{adv}} = \theta_{\text{delay}} = 2GM/(r_0\,c^2)$	Lensing-angle decomposition: equal contributions from transverse advection by inflow and asymmetric coordinate-time delay.	Split the photon trajectory integral into two terms: one from the transverse component of inflow, one from coordinate-time asymmetry near the mass.	*P3 §2.1*

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Derived Math	Description	Derivation	Source
▹ $\mathbf{B}_g(\mathbf{r}) = \frac{G}{c^2 r^3}[3(\mathbf{J}\cdot\hat{\mathbf{r}})\,\hat{\mathbf{r}} - \mathbf{J}]$	Gravitomagnetic dipole field of a body with angular momentum $\mathbf{J}$; identical structure to a magnetic dipole.	Integrate the mass-current distribution of a rotating body using the magnetic-dipole analog with v^2/c^2 coupling.	*P3 §1.7*

9. Black-Hole & Horizon Physics

The Painlevé–Gullstrand acoustic metric reproduces Schwarzschild geometry, Hawking radiation, and frame dragging with no fitted parameters.

Derived Math	Description	Derivation	Source
$ds^2 = (c^2 - v_{\text{in}}^2)\,dt^2 - 2\,v_{\text{in}}\,dr\,dt - dr^2 - r^2\,d\Omega^2$	**Painlevé–Gullstrand acoustic metric** for waves in inflowing Aether; equivalent to Schwarzschild geometry.	Write the line element seen by a wave with constant speed c in the rest frame of an Aether medium that itself flows inward at v_{in}.	*P3 §7*
$\Omega_{\text{FD}} = \frac{G\,J_\oplus}{2\,c^2\,R^3}\cos\delta = 39.3\ \text{mas/yr}$	**Gravity Probe-B frame-dragging rate** for Earth's angular momentum at the GP-B orbit; no fit parameters.	Evaluate the gravitomagnetic dipole field for Earth's $J_\oplus$ at the GP-B orbital radius and latitude; gyroscope rate is half this field.	*P3 §1.7*
▹ $\Omega_{\text{FD}} = \frac{1}{2}\,\mathbf{B}_g$	Larmor relation between gravitomagnetic field and gyroscope precession rate.	Apply the universal Larmor coupling: a precessing spin sees an effective field equal to half the imposed rotation of the medium.	*P3 §1.7*

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Derived Math	Description	Derivation	Source
$\left\lvert \frac{dv_{\text{in}}}{dr} \right\rvert_{r_s} = \frac{c^3}{4GM}$	**Inflow-velocity gradient** at the sonic horizon: surface-gravity scale that sets the Hawking temperature.	Differentiate $v_{\text{in}} = \sqrt{2GM/r}$ with respect to r and evaluate at $r_s = 2GM/c^2$.	*P6 §2.1*
▹ $v_{\text{in}}(r_s) = c$	Sonic-horizon condition: inflow reaches wave speed at the Schwarzschild radius.	Plug $r = r_s = 2GM/c^2$ into the inflow profile $v_{\text{in}} = \sqrt{2GM/r}$.	*P2 §1.4; P3 §7*
▹ $T_H \propto 1/M$	Inverse-mass scaling of Hawking temperature: heavier holes radiate colder, matching the GR prediction.	Combine the surface-gravity formula $T_H = \hbar\kappa/(2\pi k_B c)$ with the inflow gradient $\kappa \propto 1/M$.	*P6 §2.1*

10. Cosmological Evolution

Two-fluid thermodynamics and roton freeze-out drive the Hubble tension, the KBC void enhancement, dark energy's $w(z)$ history, and galaxy-scale flat rotation.

Derived Math	Description	Derivation	Source
$w(z)$: phantom $\to$ quintessence at $z \gtrsim 1$	**DESI equation-of-state crossover** from $w < -1$ to $w > -1$ as rotons freeze out and viscous heating shuts off.	Track $w(z)$ as rotons freeze out across the cooling history; viscous heating drops to zero and the equation of state flips sign.	*P6a §4.2*
▹ $f_n(z) = \exp\left[\frac{\alpha z}{1+z}\right]$, $\alpha = \Delta/(k_B T_0)$	Roton normal-fraction evolution with redshift; controls the dark-energy heating rate.	Integrate the Boltzmann factor $\exp(-\Delta/k_B T)$ along the cooling history, parameterized in z.	*P6a §3.1*

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Derived Math	**Description**	**Derivation**	**Source**
$K_A(T) = K_0\,[\,1 - \eta_K\,(T - T_{\rm ref})\,]$	**Temperature-dependent bulk modulus**: stiffness decreases with temperature, in analogy with superfluid helium-4.	Linearize the bulk-modulus temperature dependence around $T_{\rm ref}$, by analogy with first-sound measurements in liquid helium-4.	*P6b §2.3*
▹ $\eta_K \sim 10^{-4}\ \mathrm{K}^{-1}$	Fractional K_A change per kelvin; sensitivity coefficient calibrated to dark-matter and dark-energy amplitudes.	Fit η_K so the K_A variation simultaneously matches galactic rotation curves and the cosmic dark-energy density.	*P6b §2.3*
$H_{\rm loc} = H_{\rm avg}\sqrt{\Omega_m(1-\delta) + \Omega_\Lambda(1+\eta)}$	**Local Hubble enhancement** inside the KBC void from elevated thermal pressure of the void Aether.	Apply the Friedmann equation locally inside the KBC void, with elevated dark-energy density and reduced matter density.	*P6 §1.7*
▹ $T_{\rm void}/T_{\rm wall} \approx 2.5$	Void/wall temperature contrast from thermal-energy conservation across volume-fraction redistribution.	Conserve total thermal energy as it redistributes from $\sim 30\%$ wall volume to $\sim 70\%$ void volume.	*P6 §1.7*
$T(r) = T_{\rm center} - \dfrac{2\,\rho_A\,v_{\rm flat}^2}{K_0\,\eta_K}\ \ln(r/r_{\rm core})$	**Halo temperature profile** (logarithmic) producing flat galactic rotation curves.	Solve the steady-state equation balancing viscous heating in the inflowing halo Aether against Hubble dilution; profile is logarithmic in r.	*P6b §3.6*

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Derived Math	Description	Derivation	Source
▹ $\Delta K_A/K_A \approx 5.7\times10^{-6}$	Bulk-modulus variation across a galactic halo needed for flat rotation; sets the galactic η_K calibration.	Require the two-fluid orbital equation $v^2 = GM/r + (r/2)\,dc_1^2/dr$ to flatten across the halo; extract the needed ΔK_A.	*P6b §3.3*

11. Quantum-Mechanical Emergence

Quantum dynamics, the electron rest mass, and the Coulomb constant all emerge from the same ring-vortex / Bjerknes machinery acting on the Aether condensate.

Derived Math	Description	Derivation	Source
$\psi = \sqrt{\rho/m_A}\,e^{i\phi} \;\Rightarrow$ $i\hbar\,\partial_t\psi = \hat{H}\psi$	**Schrödinger equation emergence** via the Madelung transformation of the condensate, augmented by the Wallstrom quantization condition.	Rewrite the condensate field as $\psi = \sqrt{\rho/m_A}\,e^{i\phi}$; the continuity and Euler equations, plus Wallstrom single-valuedness, reassemble into Schrödinger.	*P7 §4.10*
$\kappa_e = h/(2m_A) = \pi\hbar/m_A$	**Quantum of circulation** for the electron vortex ring; winding number one and atomic-mass denominator.	Require the wavefunction to be single-valued around a closed loop; the minimum circulation is $h/(2m_A)$ for unit winding.	*P7a §3.1*
$m_e c^2 = \frac{1}{2}\rho_A\,\kappa_e^2\,r_{\text{loop}}\left[\ln\left(\frac{8\,r_{\text{loop}}}{\xi}\right)-\frac{1}{2}\right]$	**Donnelly kinetic-energy formula** for a thin vortex ring; the bare baseline for the electron rest mass.	Integrate the kinetic-energy density of the irrotational circulating flow around the vortex; cut the logarithmic divergence at the healing length.	*P7a §1*

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Derived Math	Description	Derivation	Source
$k = \dfrac{\pi\,\rho_A\,\hbar^2\,A^2}{8\,m_e^2\,e^2}$	**Coulomb constant in amplitude form**: secondary Bjerknes force from axial oscillation of two rings with amplitude A.	Apply Bjerknes' two-monopole formula with each ring treated as a piston of area πr_{loop}^2 oscillating at the Compton frequency with amplitude A.	*P7a §11.3*
▹ $P(r) = P_\infty - \rho_A\,\kappa_e^2/(8\pi^2\,r^2)$	Bernoulli pressure distribution inside an irrotational vortex; drives material toward the core.	Apply Bernoulli's equation along the irrotational azimuthal flow; faster flow at smaller r produces lower pressure.	*P7a §3.2*
▹ $\delta\rho(r)/\rho_A \approx -\kappa_e^2/(8\pi^2\,c^2\,r^2)$	Density deficit from the Bernoulli pressure drop; venturi effect inside the vortex.	Convert the Bernoulli pressure drop into a density deficit via the linearized equation of state $\delta P/\rho \approx c^2\,\delta\rho/\rho$.	*P7a §4.2*
▹ $S_{\text{eff}} = \pi\,r_{\text{loop}}^2$	Effective piston area of the ring's axial oscillation; sets the monopole source amplitude in the Bjerknes formula.	Identify the disk of radius r_{loop} bounded by the ring as the geometric piston area swept by axial motion.	*P7a §11.2*
▹ $X_{\text{core}} = \rho_{\text{core}}/\rho_A \approx 4$–$5$	Core compression ratio of bound material in the electron ring; remarkably modest for a charged lepton.	Divide the bound ring rest energy by the volume of material spread over the ring's circumference $2\pi r_{\text{loop}}$; convert to a density ratio.	*P7a §6.1*
▹ $f_{\text{core}} \approx 5$–$14\%$	Core-energy fraction reconciling the bare Donnelly KE with the observed electron rest energy.	Subtract the Bernoulli-corrected Donnelly KE from $m_e c^2$ and divide by $m_e c^2$; the remainder is f_{core}.	*P7a §5.2*

Appendix: Symbol Glossary

Symbol	Meaning
K_A	Aether bulk modulus (elastic stiffness)
ρ_A	Aether rest density
m_A	Aether atomic mass (constituent quantum)
ε	Binding-cycle asymmetry fraction ($\sim 10^{-42}$)
c	Wave speed in the Aether, $\sqrt{K_A/\rho_A}$
ξ	Healing length, $\hbar/(m_A c\sqrt{2})$
r_{loop}	Electron ring major radius $\approx ƛ_C$
$ƛ_C$	Reduced Compton wavelength, $\hbar/(m_e c)$
v_{in}	Aether inflow velocity toward a mass
r_s	Schwarzschild / sonic-horizon radius
κ_e	Quantum of circulation, $h/(2m_A)$
ψ	Condensate amplitude / order parameter
σ	QCD string tension, ≈ 1 GeV/fm
X	Bridge density-compression factor
f_{core}	Bound-core mass fraction of m_e (5–14%)
K_C	Combined parameter $\rho_A/m_A^2 \approx 7.085 \times 10^{65}$
η_K	Temperature sensitivity of K_A, $\sim 10^{-4}$ K^{-1}
α_G	Gravitational fine-structure constant, $Gm_e^2/(\hbar c)$
T_H	Hawking temperature of a black hole
Ω_{FD}	Frame-dragging precession rate
$w(z)$	Dark-energy equation-of-state parameter
$f_n(z)$	Normal-component fraction vs. redshift
ζ_n	Bulk viscosity of the normal component
c_1	First-sound speed in the two-fluid Aether